Solutions



## SAC 2: Rates and Equilibrium Name: _____

## Teacher: MC,RH, SP, NP, JJ ( please circle)

**Section A. For the following questions, circle the letter corresponding to the best alternative.**

**Question 1**

Catalysts are thought to increase the rate of a chemical reaction by

**A.** supplying the energy needed to overcome the activation energy.
**B.** increasing the temperature of the reaction.
**C.** providing less surface area for the reaction.
**D.** providing an alternative reaction pathway with a lower activation energy.

**Question 2**

100 mL of 1.00 M HCl is added to a 2 g piece of limestone, $CaCO_3$
Which of the following will **not** increase the initial rate of reaction?

**A.** Adding 150 mL of 1.00 M HCl in place of 100 mL of 1.00 M HCl
**B.** Adding 100 mL of 2.00 M HCl in place of 100 mL of 1.00 M HCl
**C.** Heating the 100 mL of 1 M HCl before adding it to the limestone.
**D.** Adding 100 mL of 1.00 M HCl to the powdered $CaCO_3$ in place of a single piece of limestone.

**Question 3**

The rates of many chemical reactions double for a ten degree rise in temperature.
An increase in which of the following factors may contribute to this rate increase.

I. The number of collisions occurring in a given time
II. The energy of each collision
III. The activation energy

**A.** I only
**B.** II only
**C.** III only
**D.** Both I and II

## Question 4

The value of the equilibrium constant for the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ is 0.25 at 440°C.

What would the value of the equilibrium constant be for the following reaction at the same temperature?

$$\tfrac{1}{2} H_2(g) + \tfrac{1}{2} I_2(g) \rightleftharpoons HI(g)$$

$\sqrt{\frac{1}{K_1}} = \sqrt{\frac{1}{0.25}} = 2$

A. 0.25

B. 0.50

C. 2.0

D. 4.0

## Question 5

The expression for the equilibrium constant for a reaction is

$$K_c = \frac{[B][C]}{[A]^2}$$

At a certain temperature the values of [A], [B] and [C] are all 0.20 mol $L^{-1}$. What happens to the value of $K_c$ when all three values are doubled to 0.40 mol $L^{-1}$?

**A.** It is halved.

**B.** It does not change.

**C.** It doubles.

**D.** It increases by a factor of four.

**The next two questions relate to the following information**

Sulfuric acid, $H_2SO_4$, is produced industrially by the Contact Process. The main reaction involves the following equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197$ kJ mol$^{-1}$

**Question 6**

A chemist investigating this equilibrium in the laboratory has a 2.0 L vessel at equilibrium at 200 °C. The vessel is found to contain 0.020 mol of $SO_2(g)$, 0.010 mol of $O_2(g)$ and 0.040 mol of $SO_3$ (g).

The equilibrium constant for this reaction under these conditions is

**A.** $1.25 \times 10^{-3}$ M

**B.** $2.50 \times 10^{-3}$ M

**C.** 400 $M^{-1}$

**D.** 800 $M^{-1}$

**Question 7**

The pressure in the converter where the equilibrium reaction above takes place, is maintained at approximately 1 atmosphere. A higher pressure would result in a

**A.** higher equilibrium yield of $SO_3$ and a higher rate of reaction

**B.** higher equilibrium yield of $SO_3$ and a lower rate of reaction

**C.** lower equilibrium yield of $SO_3$ and a higher rate of reaction

**D.** lower equilibrium yield of $SO_3$ and a lower rate of reaction

**Questions 8 & 9 refer to the following information**

Two saturated aqueous solutions of calcium hydroxide were prepared in different flasks using in each case a slight excess of calcium hydroxide so that some remained undissolved. The system in each case is represented by the equilibrium

$$Ca(OH)_2\ (s) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$$

The solutions were then used in the two experiments described below.

**Question 8**

A very small volume of 10 M hydrochloric acid was added to one flask. Which of the following observations is most likely?

**A.** no change is observed

**B.** calcium chloride is precipated.

**C.** more calcium hydroxide precipitates

**D.** more calcium hydroxide dissolves

**Question 9**

A small volume of concentrated sodium hydroxide solution was added to the second flask. Which of the following observations is most likely?

**A.** No change is observed

**B.** More calcium hydroxide is precipitated

**C.** Some solid calcium hydroxide dissolves

**D.** A precipitate of NaOH(s) forms

**Questions 10 and 11 refer to the following information.**

A mixture of ethene gas and hydrogen gas was allowed to react in a sealed, 500 mL flask. The equilibrium that resulted may be represented as

$$C_2H_4(g) + H_2(g) \rightleftharpoons C_2H_6(g)$$

At equilibrium, there was found to be $1.25 \times 10^{-1}$ mol of $C_2H_6(g)$, $7.50 \times 10^{-2}$ mol of $C_2H_4(g)$ and $8.35 \times 10^{-2}$ mol of $H_2(g)$ present in the flask.

**Question 10**

The equilibrium constant for this reaction under these conditions is

**A.** 0.0501 $M^{-1}$
**B.** 0.100 $M^{-1}$
**C.** 9.98 $M^{-1}$
**D.** 20.0 $M^{-1}$

**Question 11**

Which one of the following would result in a change in the value of the equilibrium constant for this reaction?

**A.** Increasing the concentration of $C_2H_4$ (g)
**B.** Adding a catalyst.
**C.** Decreasing the temperature of the reaction flask.
**D.** Increasing the pressure in the flask by adding an inert gas

**Questions 12 and 13 refer to the following information**

The normal pH of human blood is between 7.38 and 7.43 and pH values outside this range can be fatal. The following equilibrium system is involved in maintaining blood pH within this range:

$$CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

**Question 12**

If carbon dioxide levels were to rise (such as in the case of a pneumonia sufferer) this would

**A.** cause a decrease in the concentration of the $HCO_3^-$ ion.

**B.** cause the blood pH to fall.

**C.** cause an excess of water in the blood and surrounding tissues.

**D.** cause the blood pH to rise.

**Question 13**

In a press release, a pharmaceutical company states that it has developed a drug which acts as a catalyst in the above system. It is claimed that the drug works by "causing an equilibrium shift to the right, thus reducing the amount of $CO_2$ in the blood of pneumonia sufferers".
Choose from the following the best statement about this claim.

**A.** It is likely to be true because a catalyst always increases the rate of the forward reaction.

**B.** It is likely to be true because catalysts work best when the amount of $CO_2$ present is high.

**C.** It cannot be true because a catalyst does not change the position of the equilibrium.

**D.** It cannot be true because it would actually cause a shift to the left, thus increasing the amount of $CO_2$ in the blood.

**END OF SECTION A**

**Section B** **Answer all question**

**Question 1**

A mixture of hydrogen gas and iodine gas is injected into a vessel that is then sealed. The mixture will establish an equilibrium system as described by the following equation.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

a. In an experiment, 4.00 mol of iodine and 2.00 mol of hydrogen were added to a 5.00 L reaction vessel. The amount of iodine present at equilibrium was 2.82 mol. A constant temperature was maintained in the reaction vessel throughout the experiment. All 3SF.

i. Write the expression for the equilibrium constant for this reaction.

$$K = \frac{[HI]^2}{[H_2][I_2]}$$

1 mark

ii. Determine the equilibrium concentrations of hydrogen, iodine, and hydrogen iodide, and calculate the equilibrium constant. 5.00L

| | $H_2$ | $I_2$ | $2HI$ |
|---|---|---|---|
| n | 2.00mol | 4.00mol | 0.00mol |
| | −1.18mol | −1.18mol | 2×1.18 |
| n at eq | 0.82mol | 2.82mol * | 2.36mol * |
| [ ] at eq ÷ 5.00L * | 0.164M | 0.564M | 0.472M |
| | [H₂] | [I₂] | [HI] |

$$K = \frac{(0.472M)^2}{(0.164M)(0.564M)} = 2.41$$ *

No mark if unit given

4 marks

−1 if not 3SFs

5

b. A graph of the decrease in the amount (in mol) of $I_2$ until after equilibrium is reached is shown in Figure 1.

i. Add clearly labelled graphs to Figure 1, to show how the amounts of $H_2$ and HI changed over the same period of time. A reasonable attempt to draw the graphs to scale is required. (2 marks)

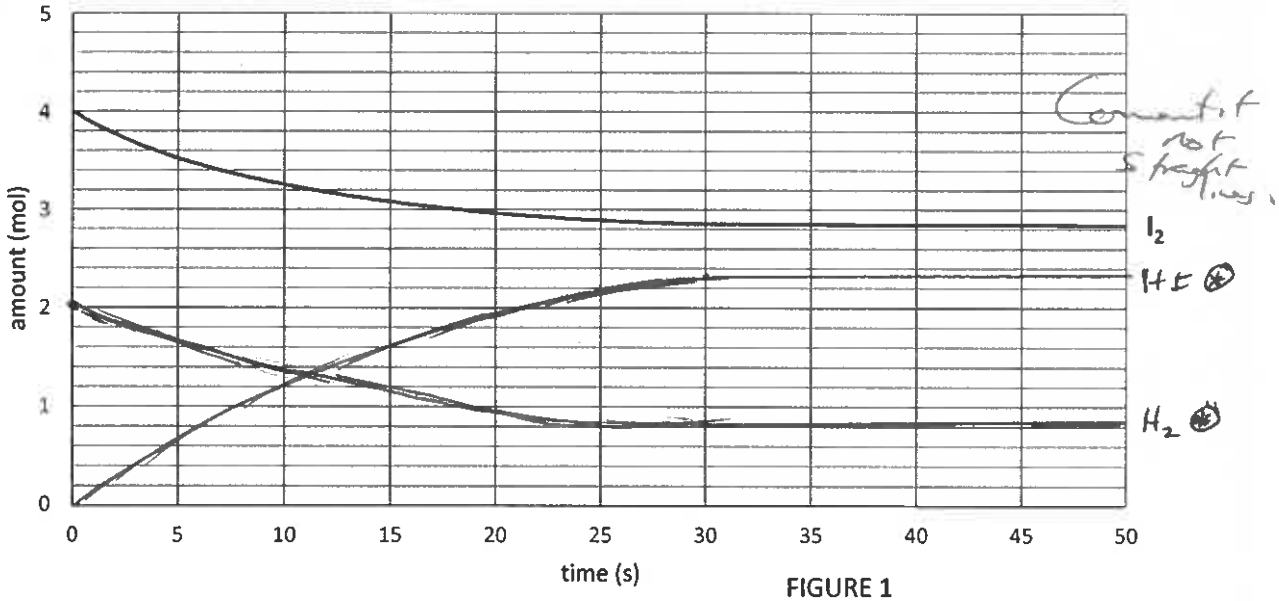


FIGURE 1

ii. Indicate on Figure 2 below, how the change in the amount of $I_2$ (only) would have been different if a catalyst had been present from the beginning. Assume all other conditions remain the same. (2 marks)

(*) Earlier to reach equilibri
(*) Same amount at equilibrium.

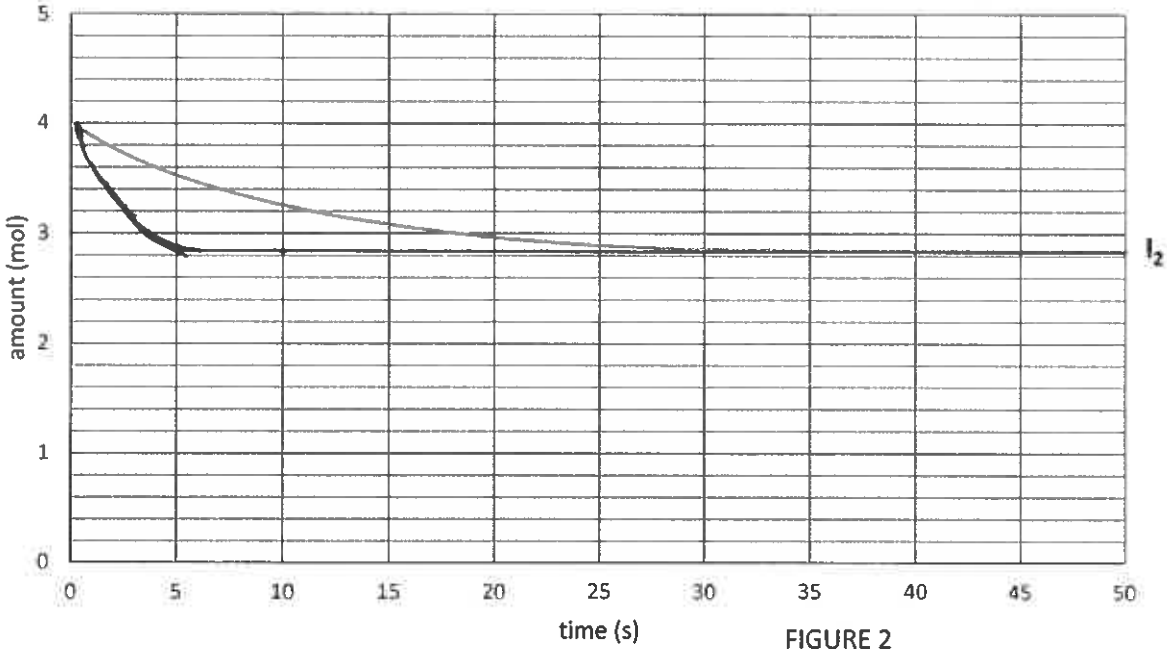


FIGURE 2

(4)

c.The forward reaction

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g) \qquad \Delta H = -ve$$

Colourless colourless purple

is an **exothermic** process. Complete the table below showing, for each of the indicated changes, the ***overall*** change in the mixture when equilibrium is re-established, ***compared to the equilibrium mixture before the change.***

| | Change to the system | Direction of reaction (forward; backward or neither) | Overall change to the intensity of the purple colour (increased, decreased, remains the same) | Overall change to rate of forward reaction (increased, decreased, remains the same) |
|---|---|---|---|---|
| i. | Temperature is decreased | Forward | Increased Purple | Decreased |
| ii. | Volume is decreased | Neither | Increased | Increased |
| iii. | Hydrogen gas is added (at constant volume) | Backward | Decreased | This cell to be left blank |

8 marks

**Total 17 marks**

8

**Question 2**

When chlorine is dissolved into cold, dilute sodium hydroxide, the following equilibrium is established

$$\overset{0}{Cl_2}(aq) + 2OH^-(aq) \rightleftharpoons \overset{-1}{Cl^-}(aq) + \overset{+1}{ClO^-}(aq) + H_2O(l)$$

**a.** Use oxidation numbers to clearly demonstrate why this is a redox reaction.

$Cl_2$ has oxidation number of 0 but in the products it has changed. now there is $Cl^-$ (oxidation number −1) and $ClO^-$ (chlorine's oxidation number is +1)

To get 2 marks either 1 number change & say oxid or red 2 marks
OR 2 marks for 2 sets of number changes

**b.** Identify the reducing agent in the **reverse** direction of the equilibrium ie in the reverse reaction. Explain your answer.

Oxidized ∴ oxidation number increases.

So the reducing agent is $Cl^-$ as its oxidation number increases from −1 to 0 (in $Cl_2$)

Any reasonable explanation for 2nd mark. 2 marks

* Identify $Cl^-$
* explanation (probably using oxidation number increase)

**c.** If more $Cl_2$ was added to the equilibrium mixture, what would happen to the pH of the mixture. Explain your answer.

If $Cl_2$ is added the system would partially oppose this and consume some $Cl_2$ & shift to the right. This means $OH^-$ will be consumed and so pH would decrease.*

* For explanation using LCP or collisions. 2 marks

**Total 6 marks**

6

**Question 3**

For people who suffer from bronchitis, even low concentrations of ozone, $O_3$, irritate the lining of the throat and can cause headaches. $NO_2(g)$ from car exhausts reacts with oxygen as follows

$$O_2(g) + NO_2(g) \rightleftharpoons NO(g) + O_3(g)$$

Car exhaust fumes also contain volatile organic compounds (VOC). These react with NO gas.

Explain, using Le Chatelier's principle, how a rise in VOC concentration will change the ozone concentration.

If concentration of VOC increase they will react with NO and reduce NO. The equilibrium system will partially oppose the decrease in NO and shift to the right. Hence more $O_3$ will be produced also.

**2 marks**

For LCP explanation.

2 points for full mark

2

**Question 4**

For the reaction $2\,X(g) + Y(g) \rightleftharpoons 2\,Z(g)$

The equilibrium constant, $K_c$, is equal to 4.0 at a particular temperature.
2.0 mol of X, 2.0 mol of Y and 2.0 mol of Z are introduced into a 1.0 L flask.

**a.** Write an expression for the equilibrium constant, $K_c$

$K = \frac{[Z]^2}{[X]^2[Y]}$

Not molecular

1 mark

**b.** Give the unit for $K_c$. $M^{-1}$ Must be capital

1 mark

Units but not positive

**c.** Prove that the mixture is not at equilibrium.

All conc are initially 2.0M $Q = \frac{(2.0M)^2}{(2.0M)^2(2.0M)} = 0.50M^{-1}$ (*)

$0.50M^{-1}$ is less than K of $2.0M^{-1}$ and so equilibrium has not yet been achieved (*)

2 marks

**d.** Determine the volume of the container for this particular mixture to be at equilibrium.

As $[Z]^2 = [X^2]$ at whatever Volume it all depends on [Y]

$\frac{1}{[Y]} = 4.0M^{-1}$ $\therefore [Y] = \frac{1}{4.0} = 0.250M$

~~$V = n \div [Y] =$~~ For $[Y] = 0.250M$

$C = \frac{n}{V} = \frac{2.00\,mol}{8.00L} = 0.250M$ $V = 8.0L$ *

2 marks

*Any logical working/ explanation

**e.** If the volume of the equilibrium mixture was suddenly doubled, what effect would this have on overall the amount, in mol, of reactant Y and the overall concentration of reactant Y? Explain your answer using Le Chatelier's principle.

If volume is increased ~~the sys~~ all concentrations will decrease (Pressure also decreases) So the system partially opposes the decrease in pressure + shifts to the left (more particles) to increase pressure. (*) Hence there will be more mol of Y but as it is only partially opposing the decrease in [Y] the [Y] will overall be less (*) than at the start.

(*) Explanation with LCP

3 marks

**Total 9 marks**

**END OF SAC**
**Total Marks 47**

9